

PokeDotz: a fun, immersive and interactive mathematical sandbox with rich feature numbers representations in virtual reality

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ABSTRACT

This work presents a visual and interactive mathematical sandbox prototype created in a Virtual Reality (VR) environment with the main goal of exploring potentialities to visualize rich features numbers representations from the perspective of a mathematical savant and, also, to manipulate them through math operations in an intuitive, fun and immersive way. The starting point for creating this experience was the opportunity to apply 3D user interfaces (3DUI) in a VR model able to bring feelings and perceptions of a math savant to everyone. As explained by Daniel Tammet, in a TED Talks, the world of words and numbers that a mathematical savant sees is full of colors, shapes, emotions and personality. It means, it's an experience more focused on aesthetics and the synesthesia with the numbers than on the abstract reasoning of math formulas. The VR set-up was composed by a head-mounted display (HMD), controllers and trackers, allowing the user to visualize little monster representations of the numbers (the PokeDotz) and, also, to interact with them by selecting and combining the PokeDotz using math operations devices with 3D interactions. In short, when a user manipulates the PokeDotz with mathematical operations (sum, subtract, divide or multiply), it is possible to visualize results for each combination by creating new aesthetics for the monsters (shape, scale, color, texture, behavior and more). As preliminary results, this prototype meets the 3DUI contest requirements, with its greatest value being the visualization and interaction techniques that allow people to experience moments of empathy with the creative and fun thought patterns of a savant mathematician.

Keywords: Virtual reality, Visualization, 3DUI, Math savant, Numbers representation.

Index Terms: Human-centered computing – Visualization; Human-centered computing – Virtual reality.

1 INTRODUCTION

It is generally agreed that the Arabic numeral system is an intelligent way of representing numbers, also being fundamental for the popularization of mathematics, especially due to the limited space of writing in printed and electronic media. On the other hand, the need for abstraction capacity can constitute a few cognitive barriers for some people [2], causing a process of dissociation between the symbol (signifier) and what it does represent (signifier) [3], which would otherwise be instinctive, making difficult the learning process in mathematics.

Unlike traditional communication media, 3D user interfaces (3DUI) and the Virtual Reality (VR) present a less restrictive space, enabling the user different ways to visualize, select and manipulate numerical quantities that would never fit within the limits of a page, a display or a numeric keypad [4, 5, 6]. Furthermore, 3DUI and VR allow the user to explore more intuitive and less abstract forms of

numerical representations, including the association of sensory stimuli, such as: colors, shapes, textures, patterns and movements [7], creating a favorable scenario for speculation about what it would be like to understand numbers or math operations based on the synesthetic ability of a savant mathematician [1].

Based on those understanding, our research team considered creating an experience capable of simulating aspects of a math savant visualization through the creation of little monsters with numerical information (small dots above them – the ‘Dotz’) and aesthetics personality (colors, shapes, textures, and more – the ‘Poke’) (Figure 1). In addition to the visualization of PokeDotz, the 3DUI techniques applied in this VR system also allow the user to interact with the little monsters (numbers) by combining them through math operations based on video game mechanics, giving to the user a gamification narrative that may enhance the attention transfer process [8] and, so on, the learning of mathematics (challenges, exploration, progression, rewards, and more) [6].

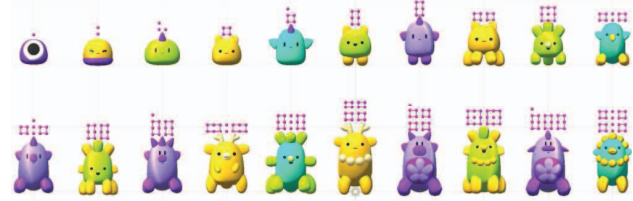


Figure 1: The PokeDotz are numbers representations in dots and in shape, color, scale or face expressions.

2 SYSTEM DESIGN AND INTERACTIONS PRINCIPLES

To create the VR prototype of a visual and interactive mathematical sandbox with rich feature numbers representation that allows the user to go beyond traditional perception boundaries of seeing, manipulating and understanding the numbers – similar perception of a math savant – five steps were emerged: a) sensorial and cognitive feature rich numbers representations; b) memory palace (interface organization); c) selection, manipulation and math operations; and, e) gamification narrative.

2.1 SENSORIAL AND COGNITIVE FEATURE RICH NUMBERS REPRESENTATIONS

In essence, the system allows the user to visualize the aesthetics of numbers represented by little monsters which we're calling PokeDotz. "I believe our personal perception to see is the heart of how we acquire knowledge. Aesthetics judgments rather than abstract reasoning, guide and shape the process by which we all come to know what we know." [1]. Based on that explanation of a savant mathematician, our research team designed to each number a representation, providing additional information about them, such as: shape, scale, color, emotions, behavior, and more. It is important to underline that sensorial and cognitive feature rich numbers representations were an imperative work to this project, because the necessity of creating to each number a distinct characterization (face, color, size, personality and relationship with other numbers).

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Consequently, the user can make use of human pattern recognition capacities to fill numbers with extra math information, establishing multiple relationships and associations among them [4].

Each PokeDotz has its magnitude represented visually in multiple ways. For example, in addition to visualizing the aesthetic characteristics of the little monsters, the representation system for each number presents small spheres of dots (Dotz) on the little monsters (Poke) to represent quantities and grouping according to their divisor [5]. So, the visualization of its properties becomes immediate and instinctive, like prime numbers (e.g., featured with purple color and with an isolated sphere dot on top). In another example, it's easy to identify numerical relationships by similarity patterns (e.g., $12 = 4 \times 3 = 9 + 3$).

2.2 MEMORY PALACE (INTERFACE ORGANIZATION)

The memory palace was a useful theory to help organize the system interface in a logical way that may bring synesthetic references to the user learning about mathematics [2]. Based on a 12-spiral design, the initial interface presents a stage where the user can easily identify where the PokeDotz were located on the 3D space (Figure 2). They're organized according to their divisors, which are represented in different color lines, reinforcing the aesthetic and synesthetic references to the numbers and to the possibility of combining them in mathematical operations just by identifying characteristics, attributes or correlations by their location [2].

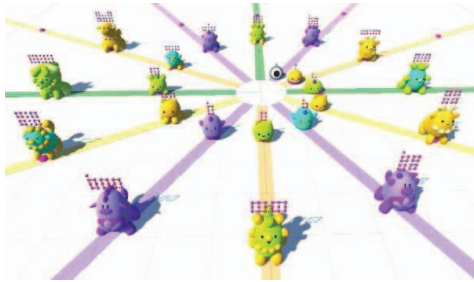


Figure 2: 12-spiral interface organizes the PokeDotz.

2.3 SELECTION, MANIPULATION AND MATH OPERATIONS

About interactions and object manipulations, the system allows the user to select any PokeDotz by pointing and pulling them with a VR controller. However, in order to make the selection easier to see, the selected PokeDotz float above the spiral stage allowing the user to visualize one, two or more little monsters to create a combination among them. It means, the user simply needs to point at the desired number and press the controller trigger to grab and lift it to a kind of virtual workbench.

When selecting PokeDotz, the user has the freedom to manipulate them using math operations in an intuitive, fun and immersive way. For such, a calculating machine can be activated using the second trigger button of the Quest 2 device, while the other buttons may select the desired operation (sum, subtract, divide or multiply) (Figure 3). This play-based learning process encourages the user to carry out math operations based on elementary arithmetic [6], but visualizing intuitive information about the numbers. After all, the arithmetic operations result unlock new monster aesthetics, while the user learns the relationship between them (e.g., $\text{PokeDotz } 1 + \text{PokeDotz } 2 = \text{PokeDotz } 3$) [6].



Figure 3: Manipulating PokeDotz with the mathematical machine.

2.4 GAMIFICATION NARRATIVE

As a final element of the system design and interaction principles, the use of gamification narratives, as progression, discovery, game loop and currency, were applied to create an attention transfer meaning to the user [8].

In particular, the game mechanics of progression, discovery and exploration were applied when the user is playing the selection, manipulation and mathematical operations of the PokeDotz. After all, using the mathematical machine device in the virtual workbench for the combination of the very essence of the little monsters (numbers) was considered an act of high transfer attention level in a game task [7, 8].

Also, as the basis for the plot of the PokeDotz combining experience, the system starts with just the sum operation in the mathematical machine device. Thus, to unlock new mathematical operations (subtract, divide or multiple), the user needs to buy them with 'stars currency' [4]. In this way, each PokeDotz discovered with a basic arithmetic operation generates one 'star'. The more the user plays with the possible combinations between the little monsters, the more 'stars' will be received as a reward.

3 RESULTS AND FUTURE WORK

This work presented the design and development of a mathematical sandbox prototype created with 3DUI and VR techniques with the main goal of encourage the user to explore feelings and perceptions of a mathematical savant with numbers and math operations. Based on rich feature numbers representations and memory palace in a 3D environment the system presents the concept of PokeDotz (little monsters representing the numbers with colors, shapes, textures, scales, location, dots, and more), allowing the user to conduct 3D interactions by combining them based on elementary arithmetic.

Even though user testing has not yet been carried out, the results perceived by the research team of this project indicate paths for possible ways of exploring learning with numbers in a visual, immersive, fun and interactive way. By the date of the 3DUI contest event, we will have already conducted user tests to verify aspects of usability, learning process and user experience with the system. Also, as future work it is expected to include more gamification elements to the plot, in order to enhance the user attention's transfer to the learning process of mathematics through VR.

Given the requirements of the 3DUI contest, the proposed system meets the requirements, with a greater emphasis on the criteria of novelty in design and enjoyment/engagement (fun) in immersive VR platforms for learning mathematics and social empathy.

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